Priority-Aware Task Offloading and 3D Trajectory Optimization in NOMA-Enabled Multi-UAV MEC-IoT Networks Using MAPPO

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***Abstract*—Integrating unmanned aerial vehicles (UAVs) with mobile edge computing (MEC) establishes an agile paradigm for provisioning compute capabilities to energy-constrained Internet of Things (IoT) nodes in dynamic, infrastructure-deficient environments. Nevertheless, prevailing computation offloading architectures conventionally treat workloads uniformly, failing to accommodate the heterogeneous urgency, delay tolerances, and operational criticality inherent to real-world sensory deployments. Furthermore, simultaneously steering multi-UAV 3D trajectories, managing non-orthogonal multiple access (NOMA) uplink interference, and scheduling priority-differentiated tasks under strict aerial propulsion limits constitutes a high-dimensional, tightly coupled mixed-integer non-convex challenge. To address these bottlenecks, this paper proposes the *Priority-Aware Multi-UAV Joint Optimization (PA-MUJO)* framework. We formulate the joint problem as a stochastic mixed-integer nonlinear program (MINLP) and devise a multi-agent proximal policy optimization (MAPPO) scheme operating under the centralized training with decentralized execution (CTDE) architecture. A composite priority characterization is established to dynamically weigh task urgency against application criticality, enabling the aerial fleet to preferentially schedule and compute high-priority requests. To mitigate reinforcement learning action complexity, NOMA uplink transmit powers are determined analytically via bisection search, decoupling physical transmission control from actor policy gradient updates. Extensive simulations demonstrate that PA-MUJO considerably curtails priority-penalized execution latency and economizes UAV propulsion energy relative to state-of-the-art benchmarks, exhibiting resilient multi-agent spatial coordination.**

***Index Terms*—Unmanned aerial vehicles (UAVs), mobile edge computing (MEC), Internet of Things (IoT), multi-agent reinforcement learning (MARL), non-orthogonal multiple access (NOMA), priority-aware offloading, 3D trajectory design.**

I. Introduction

THE proliferation of the Internet of Things (IoT) has accelerated the deployment of autonomous sensory and embedded devices across smart transportation, industrial telemetry, environmental surveillance, and emergency response [1], [2]. In these applications, ground devices (GDs) continuously collect observational telemetry and produce computational workloads requiring near-instantaneous execution. However, typical GDs operate under stringent hardware limits, including miniature battery capacities and restricted processor clock frequencies. Executing data-intensive algorithms on-device rapidly depletes local energy reserves and induces severe processing latency [3], [4]. Mobile edge computing (MEC) mitigates this physical constraint by deploying computing servers at the network edge, enabling resource-limited terminals to offload demanding workloads via wireless channels [2], [5].

Despite the benefits of MEC, conventional terrestrial edge architectures rely on stationary base stations whose fixed geographic footprint renders them vulnerable to physical obstructions, severe propagation shadowing, and exorbitant deployment overhead in remote or devastated areas [6], [7]. To circumvent these physical bottlenecks, unmanned aerial servers carrying onboard edge computing payloads have emerged as an agile, on-demand paradigm [8], [9]. Leveraging high spatial maneuverability and line-of-sight (LoS) air-to-ground (A2G) propagation, a cooperative fleet of rotary-wing UAVs can dynamically relocate across three-dimensional (3D) airspace, provisioning elastic edge compute capacity and adaptive coverage directly above dispersed ground terminals [6], [10].

Nevertheless, orchestrating cooperative multi-UAV MEC networks presents three critical unresolved challenges:

1) *Workload Urgency and Priority Inversion:* Existing offloading literature predominantly presumes homogeneous workload characteristics, treating incoming tasks uniformly through first-come-first-served scheduling [4], [9], [11]. Real-world IoT deployments, however, exhibit extreme variations in operational urgency; emergency alarms and vehicular collision avoidance tolerate only millisecond-level delays, whereas periodic telemetry can tolerate longer delays [12], [13]. Indiscriminate resource allocation induces "priority inversion," where delay-tolerant background traffic monopolizes scarce aerial computing and spectral resources, resulting in fatal deadline expirations for mission-critical tasks [13], [14].
2) *3D Flight Dynamics and Propulsion Costs:* Most existing trajectory schemes restrict aerial navigation to fixed two-dimensional (2D) planes [9], [13]. While vertical elevation adjustments dynamically regulate LoS channel probabilities and overcome terrestrial shadowing [6], rotary-wing aerodynamic propulsion power varies non-linearly with airspeed, dominating onboard battery depletion by orders of magnitude over comput-

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ing circuitry [6], [15]. Balancing dynamic propagation enhancements against severe propulsion energy drains in 3D airspace under multi-UAV collision constraints remains an open challenge.

3) *Action Space Dimensionality in NOMA Uplink Power Control:* Uplink power-domain non-orthogonal multiple access (NOMA) enables multiple GDs to simultaneously offload over identical spectral bands via successive interference cancellation (SIC) [5], [16], [17]. However, directly parameterizing continuous device transmit powers inside reinforcement learning actor action spaces causes severe policy gradient variance and dimensional explosion, destabilizing multi-agent convergence.

To overcome these hurdles, we introduce PA-MUJO, a priority-aware framework for joint optimization in multi-UAV MEC networks. We formulate the joint problem of 3D aerial navigation, priority-differentiated task offloading, and NOMA uplink transmission as a stochastic mixed-integer non-linear program (MINLP) to minimize the priority-weighted aggregate latency while regulating UAV propulsion and computing energy expenditure. To resolve this combinatorial formulation, we develop a multi-agent proximal policy optimization (MAPPO) algorithm under the centralized training with decentralized execution (CTDE) architecture [8], [18]. Crucially, we establish an analytical-RL hybrid decoupling: decentralized UAV actors govern macro-level spatial movement and scheduling associations, while optimal NOMA transmit powers are determined analytically via convex bisection, guaranteeing quality-of-service (QoS) rates while stabilizing policy training.

The principal contributions of this work are summarized as follows:

- **Comprehensive Stochastic MINLP Formulation:** We establish an end-to-end multi-UAV MEC-IoT framework integrating Gauss-Markov ground device mobility, probabilistic elevation-dependent A2G channels, NOMA uplink interference, UAV edge queueing dynamics, and nonlinear rotary-wing aerodynamics under strict battery and inter-UAV collision bounds.
- **Composite Urgency-Criticality Priority Model:** We design a composite priority metric fusing real-time deadline margins with semantic application criticality. Penalizing delay proportionally to priority directs the aerial fleet to preferentially schedule and execute time-critical workloads, fundamentally eliminating priority inversion.
- **Analytical Decoupling of NOMA Power Allocation:** We derive a recursive closed-form bisection algorithm for uplink NOMA transmission powers, completely eliminating continuous power control from the RL action space and drastically accelerating policy convergence.
- **CTDE-Based MAPPO Fleet Coordination:** We design a MAPPO cooperative optimization scheme with Generalized Advantage Estimation (GAE) and clipped surrogate objectives, structuring an adaptive multi-agent policy for priority-differentiated task offloading, energy-efficient 3D navigation, and collision-free fleet coordination.

The structural layout of this article proceeds as follows: Section II examines related scholarly works. Section III details the system architecture and mathematical formulation. Section IV establishes the cooperative PA-MUJO algorithm and bisection power allocation. Section V outlines the simulation evaluation plan, and Section VI summarizes conclusions and future trajectories.

II. Related Work

A. Computation Offloading in UAV-Assisted MEC Systems

Deploying unmanned aerial vehicles (UAVs) equipped with mobile edge computing (MEC) servers has emerged as an indispensable paradigm for delivering responsive computational services to ground devices (GDs) in regions lacking ubiquitous terrestrial infrastructure [2], [7]. Early investigations in UAV-assisted computation offloading predominantly leveraged classical mathematical programming and convex optimization methodologies. Frameworks based on successive convex approximation (SCA), block coordinate descent (BCD), Lyapunov optimization, and game-theoretic matching were widely explored to co-optimize offloading decisions, CPU resource allocation, and communication parameters [4], [19]. For example, Barman et al. [4] developed an alternating optimization framework to minimize end-to-end task completion latency by coordinating offloading, caching, and server scheduling. Similarly, Sun et al. [19] proposed a two-timescale decomposition method separating long-term UAV mobility from short-term computational resource matching. Although these optimization-theoretic formulations offer valuable theoretical insights and asymptotic performance guarantees, their practical applicability in dynamic aerial networks is constrained. Specifically, they rely on idealized quasi-static channel assumptions, require frequent centralized re-computation with polynomial-to-exponential complexity, and struggle to adapt in real time to the fast-fading air-to-ground (A2G) links and stochastic user mobility characteristic of practical IoT environments [6].

To overcome the computational intractability and static assumptions of conventional solvers, deep reinforcement learning (DRL) has gained substantial traction as a model-free alternative capable of learning complex state-to-action mappings via continuous environmental interaction [20], [21]. Initial studies deployed single-agent DRL architectures, such as Deep Q-Networks (DQN) and Deep Deterministic Policy Gradient (DDPG), to optimize task offloading and UAV deployment under uncertain network conditions [3], [11]. For instance, Zhan et al. [3] integrated graph attention networks with dueling double deep Q-networks to address dependent task offloading, demonstrating the adaptability of neural network policies in capturing complex topological states. Furthermore, Zhao et al. [22] investigated collaborative offloading for directed acyclic graph (DAG) tasks in multi-UAV assisted MEC environments to minimize execution delays under battery constraints, while Xiao et al. [23] examined bi-directional computation offloading in UAV networks assisted by reconfigurable surfaces. Nevertheless, centralized and single-agent DRL paradigms encounter severe scalability bottlenecks in multi-UAV deployments: aggregating global system state information at a centralized controller incurs prohibitive signaling overhead, while the dimensionality of the joint action

space explodes exponentially with the number of aerial servers and ground devices [9].

To achieve scalable, distributed decision-making, recent research has shifted toward multi-agent reinforcement learning (MARL) frameworks [18], [24]. By employing the centralized training with decentralized execution (CTDE) architecture, multi-agent algorithms such as Multi-Agent DDPG (MADDPG) and Multi-Agent Proximal Policy Optimization (MAPPO) enable aerial nodes to learn coordinated offloading and flight policies while circumventing non-stationarity [8], [9]. In particular, Kang et al. [8] demonstrated the effectiveness of MAPPO in hierarchical aerial networks, showing that sharing a centralized state-value critic during offline training stabilizes policy learning across concurrent agents while allowing each UAV to execute autonomous offloading decisions using only local observations. Similarly, Tariq et al. [9] proposed an actor-critic MARL scheme for multi-UAV resource allocation, highlighting the superiority of decentralized execution in reducing communication signaling latency.

Despite these advancements, a critical limitation persists across the vast majority of existing computation offloading literature: workloads are treated homogeneously under the implicit assumption of uniform urgency and identical delay tolerances [3], [4], [9], [11]. Conventional schemes optimize system-wide average completion latency or aggregate energy consumption in an indiscriminate manner, failing to account for the diverse QoS requirements and criticality profiles inherent in multi-application IoT networks. In dynamic operational scenarios, this priority-agnostic design induces severe "priority inversion," where scarce aerial computing and spectrum resources are occupied by delay-tolerant background telemetry, causing mission-critical events to experience intolerable deadline violations. This fundamental gap underscores the urgent necessity for a cooperative multi-agent offloading framework that systematically incorporates heterogeneous task priorities into joint 3D trajectory steering and resource scheduling.

B. Priority-Aware Task Scheduling and Offloading

In massive IoT and sensor networks, ground devices concurrently execute heterogeneous applications with starkly divergent quality-of-service (QoS) requirements, spanning from routine periodic environmental telemetry to latency-intolerant emergency signaling and closed-loop robotic actuation [7], [12]. Treating such disparate computational demands homogeneously in edge queues inevitably precipitates head-of-line blocking and severe deadline violations for mission-critical operations. Consequently, incorporating workload differentiation and prioritization into computational offloading protocols has attracted growing research attention [12], [13]. Early efforts focused on single-UAV architectures; for example, Hu et al. [12] developed a priority-aware offloading scheme that prioritized incoming computational requests based on remaining deadline strictness. However, their analytical formulation was constrained to a single aerial server and relied on orthogonal spectrum slicing, rendering it incapable of addressing multi-UAV spatial load balancing, inter-cell interference, or spectral multiplexing over high-density user populations.

To accommodate more complex topologies, subsequent studies extended priority mechanisms to multi-server and hierarchical aerial frameworks [13], [14]. Sun et al. [13] investigated priority-driven resource allocation that sorted tasks into static priority classes, leveraging convex relaxation to optimize transmit power and CPU clock frequencies. Nonetheless, their approach depended on heuristic task admission rules that lacked adaptability under stochastic air-to-ground channel fading and time-varying device mobility. Addressing spatial coordination, Peng et al. [14] formulated a priority-aware joint trajectory and task scheduling problem for multi-UAV networks using mixed-integer non-linear programming (MINLP). Although their scheme achieved bounded completion delays for high-priority traffic, it constrained UAV flight to fixed two-dimensional (2D) planes, decoupled trajectory design from short-term radio resource allocation, and relied on orthogonal frequency-division multiple access (OFDMA), which severely caps system throughput and spectral efficiency in dense IoT environments.

More recently, multi-agent reinforcement learning (MARL) paradigms have been introduced to realize adaptive, decentralized priority-aware offloading. In particular, Zhou and Sun [25] investigated priority-aware task offloading in a hierarchical aerial access network comprising high-altitude platforms (HAPs) and low-altitude UAVs, utilizing a Multi-Agent Proximal Policy Optimization (MAPPO) framework to balance server workload fairness. Although their model demonstrated the benefits of multi-agent learning in aerial offloading, task priority was abstracted as a static scalar weight without distinguishing between dynamic temporal urgency (remaining latency margin) and intrinsic application criticality (event importance). Furthermore, their framework assumed orthogonal wireless channels and planar UAV trajectories at fixed operational altitudes, thereby overlooking the throughput gains of non-orthogonal multiple access (NOMA) and the spatial reconfigurability offered by continuous 3D aerial kinodynamics [6], [15].

Crucially, an evident void remains in the current literature: existing frameworks uniformly isolate workload prioritization from non-orthogonal channel multiplexing and continuous 3D spatial mobility. Specifically, none of the aforementioned approaches models a *composite priority metric* that simultaneously factors in dynamic deadline margins and intrinsic semantic importance, nor do they couple this priority metric with uplink NOMA power-domain multiplexing and cooperative multi-UAV 3D trajectory control. Our proposed PA-MUJO framework directly overcomes these deficiencies by formulating a composite urgency-criticality priority metric $\Omega_i[t]$, which dynamically steers NOMA user clustering, decoding order determinations, UAV spatial velocity control, and centralized multi-agent reward shaping.

C. 3D UAV Trajectory Planning and Mobility Control

Early research on unmanned aerial vehicle (UAV) deployment in wireless networks predominantly constrained aerial mobility to two-dimensional (2D) horizontal planes at a fixed, predetermined altitude [10], [15]. While fixed-altitude assumptions simplify geometric path loss derivations, they forfeit

critical spatial degrees of freedom. In realistic operational environments characterized by urban structures, topography, and terrestrial obstacles, dynamically regulating UAV altitude alters the elevation angle between the aerial server and ground terminals, directly dictating the Line-of-Sight (LoS) link probability [6]. However, 3D aerial mobility introduces intricate non-linear physical trade-offs: ascending improves the elevation angle and suppresses shadowing, but amplifies free-space path loss; conversely, descending curtails signal propagation distance but severely exacerbates the likelihood of non-line-of-sight (NLoS) blockage and ground collision risks.

In practical IoT scenarios, these propagation trade-offs are compounded by ground user mobility, where terrestrial devices follow stochastic trajectories such as the Gauss-Markov mobility model [26]. In a foundational study, Li et al. [6] investigated deep reinforcement learning-based 3D trajectory design for UAV-assisted networks serving mobile ground users under elevation-dependent probabilistic air-to-ground (A2G) channels. Their work demonstrated that continuous 3D velocity and heading control significantly outperforms discrete planar waypoint navigation, enabling the UAV to dynamically track user spatial dispersion while adapting to time-varying channel states. Furthermore, rotary-wing aerodynamic propulsion power is a highly non-linear function of instantaneous flight velocity—encompassing blade profile, induced, and parasite drag components [6], [15]. Consequently, trajectory optimization cannot treat mobility costs as trivial or linear, requiring rigorous co-design between kinematic maneuvers and onboard energy expenditure.

Extending 3D trajectory planning to cooperative multi-UAV fleets introduces challenging spatial coupling, multi-agent interference, and strict inter-UAV collision avoidance constraints [8], [27]. Classical mathematical programming techniques, such as the two-timescale decomposition investigated by Sun et al. [19], decouple macro-scale trajectory design from micro-scale resource dispatching. However, such decoupled formulations rely on heuristic approximations and static channel assumptions, failing to adapt when user spatial distributions or channel states shift abruptly. While single-agent DRL and graph attention networks have been applied to multi-UAV positioning [3], they encounter severe signaling bottlenecks and computational intractability in dense networks. To overcome these limitations, multi-agent reinforcement learning (MARL) frameworks operating under the centralized training with decentralized execution (CTDE) paradigm, such as MAPPO [8], [18], have demonstrated superior efficacy in coordinating multi-UAV flight formations while guaranteeing inter-UAV safety distances using only localized agent observations.

Nevertheless, a critical deficiency across existing 3D trajectory optimization literature is that flight maneuvers are steered predominantly by generic network objectives—such as aggregate throughput, uniform coverage radius, or total offloading volume—without consideration of task-level quality-of-service (QoS) urgencies [3], [6], [8]. In networks with heterogeneous traffic, this priority-agnostic mobility steering causes UAVs to linger near dense clusters of delay-tolerant reporting devices while mission-critical clusters suffer severe packet expirations. The proposed PA-MUJO framework directly overcomes this limitation by coupling continuous 3D aerial kinodynamics with our composite urgency-criticality priority metric $\Omega_i[t]$. By incorporating priority-weighted metrics into the MAPPO cooperative policy, UAV actors autonomously modulate their 3D velocities and altitudes to establish high-gain LoS channels toward high-priority ground clusters, simultaneously ensuring aerodynamic energy efficiency and flight safety.

D. NOMA-Enhanced Aerial Communications

Massive Internet of Things (IoT) deployments demand high-density uplink connectivity under severely congested radio frequency spectrum. Traditional orthogonal multiple access (OMA) protocols—such as OFDMA or time-division multiple access (TDMA)—partition spectral bands or time slots into mutually disjoint resource blocks, strictly capping concurrent user access by the number of orthogonal channels [5], [7]. Power-domain non-orthogonal multiple access (NOMA) overcomes this bottleneck by multiplexing multiple ground devices onto identical spectral channels within the same time slot, exploiting channel gain disparities and successive interference cancellation (SIC) at the receiver to decode superposed signals [5], [17]. Integrating NOMA into UAV-assisted mobile edge computing substantially amplifies uplink spectral efficiency and scales concurrent task offloading capacity in bandwidth-constrained environments.

The synergy between NOMA and aerial edge computing offers distinct physical-layer advantages: aerial base stations can dynamically adjust their spatial positions to engineer channel gain disparities among ground terminals, directly improving SIC decoding efficiency and system throughput [5], [16]. However, joint optimization of aerial trajectories, user clustering, and transmit power allocation results in a non-convex mixed-integer non-linear programming (MINLP) problem with strong inter-variable coupling. Early efforts deployed mathematical programming techniques, such as successive convex approximation (SCA) and geometric programming, to solve NOMA power allocation and trajectory planning iteratively [16]. Nevertheless, these iterative solvers incur substantial computational delay and require static channel state information (CSI), rendering them ill-suited for latency-critical environments with stochastic user mobility.

To enhance environmental adaptability, recent literature has applied deep reinforcement learning to co-optimize NOMA power levels and UAV flight paths [16], [17]. In typical DRL formulations, the continuous transmit powers of ground devices are concatenated directly into the actor network's continuous action space alongside aerial velocity coordinates. However, this direct parameterization leads to severe policy degradation: the continuous action dimensionality scales linearly with the number of multiplexed devices, dramatically increasing sample complexity, policy gradient variance, and training instability. Furthermore, neural policy networks struggle to strictly enforce individual maximum transmit power bounds ($p_i[t] \leq P_i^{\max}$) and minimum QoS data rate constraints ($R_{i,j}[t] \geq R_i^{\text{th}}$) without excessive reward penalty tuning.

In contrast to black-box continuous action parameterization, the optimal uplink NOMA power allocation conditioned on

a given user cluster and spatial trajectory exhibits a monotonic mathematical structure that is amenable to convex optimization. The proposed PA-MUJO framework resolves the action dimensionality dilemma by establishing an analytical-RL hybrid decoupling: decentralized MAPPO actors optimize macro-level spatial actions—specifically discrete offloading associations $\chi_{i,j}[t]$ and continuous 3D velocity vectors $\mathbf{v}_j[t]$—while the exact continuous transmit powers $p_i^*[t]$ are analytically determined at each time slot via an efficient convex bisection procedure. This architectural decoupling drastically compresses the RL action space, eliminates policy gradient variance stemming from continuous power control, and guarantees deterministic satisfaction of minimum data rate constraints across all active NOMA clusters.

III. System Model and Problem Formulation

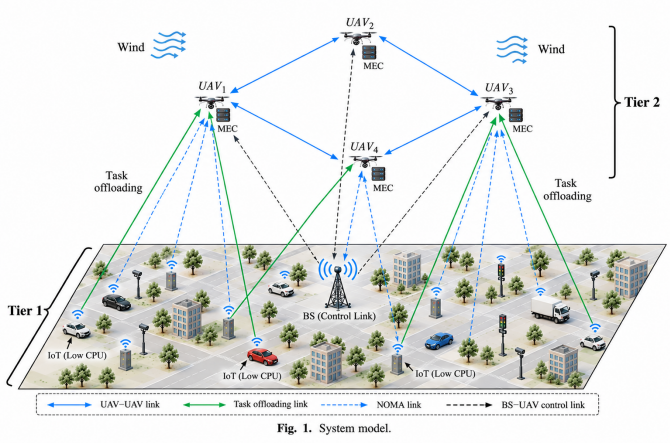


Fig. 1. System model.

Figure 1. Architecture of the proposed priority-aware multi-UAV MEC-IoT network with NOMA-enabled uplink offloading, 3D aerial trajectories, and terrestrial coordination.

A. Network Architecture and Topology

We consider an aerial-assisted mobile edge computing architecture serving a geographical operational domain $\mathcal{D} \subset \mathbb{R}^2$ over a finite operational horizon discretized into T sequential time slots, indexed by $t \in \mathcal{T} \triangleq \{1, 2, \ldots, T\}$, with uniform duration δ. The network topology is depicted in Fig. 1 and comprises:

- A set of terrestrial ground devices (GDs), indexed by $i \in \mathcal{I} \triangleq \{1, 2, \ldots, I\}$, scattered across the service area. The horizontal coordinate of device i at time slot t is denoted by $\mathbf{u}_i[t] = [x_i[t], y_i[t], 0]^\top$. Each device possesses an internal computing processor characterized by maximum CPU clock frequency f_i^{loc}.
- A fleet of rotary-wing UAVs, indexed by $j \in \mathcal{J} \triangleq \{1, 2, \ldots, J\}$, functioning as flying MEC servers. Each UAV j navigates in 3D airspace with instantaneous spatial coordinates

$$\mathbf{p}_j[t] = [x_j[t], y_j[t], z_j[t]]^\top, \quad z_j[t] \in [z_{\min}, z_{\max}], \quad (1)$$

where $z_{\min}$ and $z_{\max}$ denote the allowable lower and upper operational altitude limits, respectively, mandated by obstacle clearance and aviation safety standards. Each UAV carries an onboard edge computing server operating at CPU clock frequency f_j^{uav}.
- A terrestrial coordination station (Base Station, BS), located at fixed ground coordinates $\mathbf{w}_0 = [x_0, y_0, 0]^\top$, which serves as a central network synchronization anchor and offline training supervisor. The BS does not execute user computation tasks.

B. 3D UAV Kinematic and Fleet Mobility Model

To provide responsive edge computing over the geographical region $\mathcal{D}$, each UAV $j \in \mathcal{J}$ navigates in three-dimensional airspace with time-varying coordinates $\mathbf{p}_j[t] = [x_j[t], y_j[t], z_j[t]]^\top \in \mathbb{R}^3$ at time slot $t \in \mathcal{T}$ [6]. The vertical movement of the aerial fleet is restricted within an authorized flight envelope $z_j[t] \in [z_{\min}, z_{\max}]$, where $z_{\min}$ and $z_{\max}$ denote the minimum safety clearance altitude and maximum regulatory ceiling, respectively. Furthermore, UAV horizontal flight is bounded by the operational domain, requiring $[x_j[t], y_j[t]]^\top \in \mathcal{D}$.

Following the continuous 3D aerial kinodynamic formulations in [6] and [15], the spatial displacement of UAV j across consecutive time slots is governed by its instantaneous 3D velocity vector $\mathbf{v}_j[t] = [v_j^x[t], v_j^y[t], v_j^z[t]]^\top \in \mathbb{R}^3$. The discrete-time spatial kinematic evolution is modeled as

$$\mathbf{p}_j[t+1] = \mathbf{p}_j[t] + \mathbf{v}_j[t]\delta, \quad \forall j \in \mathcal{J}, \forall t \in \mathcal{T}, \quad (2)$$

which decomposes along the Cartesian axes into

$$x_j[t+1] = x_j[t] + v_j^x[t]\delta, \quad (3)$$
$$y_j[t+1] = y_j[t] + v_j^y[t]\delta, \quad (4)$$
$$z_j[t+1] = z_j[t] + v_j^z[t]\delta. \quad (5)$$

The instantaneous 3D flight speed is constrained by the aerodynamic performance limits of the rotorcraft:

$$v_j[t] \triangleq \|\mathbf{v}_j[t]\|_2 = \sqrt{\left(v_j^x[t]\right)^2 + \left(v_j^y[t]\right)^2 + \left(v_j^z[t]\right)^2} \leq V_{\max}, \quad (6)$$

where $V_{\max}$ is the maximum achievable flight speed. To prevent catastrophic mid-air collisions among aerial servers during cooperative fleet maneuvers, a minimum inter-UAV safety distance $d_{\min}$ must be maintained across all pairs of UAVs at every time slot:

$$\|\mathbf{p}_j[t] - \mathbf{p}_{j'}[t]\|_2 \geq d_{\min}, \quad \forall j \neq j' \in \mathcal{J}, \forall t \in \mathcal{T}. \quad (7)$$

C. Gauss-Markov Mobility Model of Ground Devices

To capture realistic mobile IoT scenarios, GDs move autonomously within region $\mathcal{D}$. Following the Gauss-Markov (GM) random mobility formulation [6], [26], temporal memory in device velocity and trajectory orientation is explicitly modeled. At each time slot $t \in \mathcal{T}$, the scalar velocity $v_i[t]$ and horizontal azimuth heading angle $\varphi_i[t]$ of device i evolve recursively according to

$$v_i[t] = \alpha_v v_i[t-1] + (1-\alpha_v)\bar{v}_i + \sqrt{1-\alpha_v^2}\,\zeta_i^v[t], \quad (8)$$
$$\varphi_i[t] = \alpha_\varphi \varphi_i[t-1] + (1-\alpha_\varphi)\bar{\varphi}_i + \sqrt{1-\alpha_\varphi^2}\,\zeta_i^\varphi[t], \quad (9)$$

where $\bar{v}_i$ and $\bar{\varphi}_i$ represent the asymptotic mean velocity and directional drift of device i, respectively. The coefficients $\alpha_v, \alpha_\varphi \in [0,1]$ parameterize the correlation memory of the Markovian process, while $\zeta_i^v[t] \sim \mathcal{N}(0,\sigma_v^2)$ and $\zeta_i^\varphi[t] \sim \mathcal{N}(0,\sigma_\varphi^2)$ represent independent Gaussian stochastic driving noises. Consequently, the 2D position vector of device i updates as

$$x_i[t] = x_i[t-1] + \delta \cdot v_i[t-1]\cos\varphi_i[t-1], \tag{10}$$

$$y_i[t] = y_i[t-1] + \delta \cdot v_i[t-1]\sin\varphi_i[t-1]. \tag{11}$$

Should an updated coordinate fall outside the boundary of $\mathcal{D}$, a specular reflection boundary condition is applied to reverse its normal heading component, maintaining all ground nodes within the designated surveillance zone.

D. Air-to-Ground Probabilistic Channel Model

The wireless links connecting ground devices to aerial UAVs are governed by elevation-dependent probabilistic air-to-ground (A2G) propagation models [6], [10]. For device $i \in \mathcal{I}$ and UAV $j \in \mathcal{J}$ at slot t, the 3D Euclidean distance is

$$\begin{aligned} d_{i,j}[t] &= \|\mathbf{p}_j[t] - \mathbf{u}_i[t]\|_2 \\ &= \sqrt{\big(x_j[t]-x_i[t]\big)^2 + \big(y_j[t]-y_i[t]\big)^2 + z_j^2[t]}, \end{aligned} \tag{12}$$

and the corresponding spatial elevation angle (in degrees) is given by

$$\psi_{i,j}[t] = \frac{180}{\pi}\arcsin\!\left(\frac{z_j[t]}{d_{i,j}[t]}\right). \tag{13}$$

The probability of establishing a line-of-sight (LoS) link depends on the elevation angle and the propagation environment:

$$\mathbb{P}_{i,j}^{\mathrm{LoS}}[t] = \frac{1}{1 + c_1 \exp\big(-c_2\big(\psi_{i,j}[t] - c_1\big)\big)}, \tag{14}$$

where c_1 and c_2 denote environment-specific empirical parameters (e.g., urban, suburban, dense urban). The complementary non-line-of-sight (NLoS) probability is

$$\mathbb{P}_{i,j}^{\mathrm{NLoS}}[t] = 1 - \mathbb{P}_{i,j}^{\mathrm{LoS}}[t]. \tag{15}$$

Let ϱ_{LoS} and ϱ_{NLoS} denote the large-scale channel attenuation coefficients for LoS and NLoS states, respectively, satisfying $\varrho_{\mathrm{LoS}} > \varrho_{\mathrm{NLoS}} > 0$. Accounting for geometric spreading with path-loss exponent α_{pl}, the effective A2G channel power gain is expressed as

$$\begin{aligned} h_{i,j}[t] &= \mathbb{P}_{i,j}^{\mathrm{LoS}}[t]\, h_{i,j}^{\mathrm{LoS}}[t] + \mathbb{P}_{i,j}^{\mathrm{NLoS}}[t]\, h_{i,j}^{\mathrm{NLoS}}[t] \\ &= \Big[\mathbb{P}_{i,j}^{\mathrm{LoS}}[t]\varrho_{\mathrm{LoS}} + \mathbb{P}_{i,j}^{\mathrm{NLoS}}[t]\varrho_{\mathrm{NLoS}}\Big]\big(d_{i,j}[t]\big)^{-\alpha_{\mathrm{pl}}}. \end{aligned} \tag{16}$$

E. NOMA-Enhanced Uplink Transmission Model

To enable multiple GDs to simultaneously offload workloads to a shared aerial server over identical bandwidth, power-domain non-orthogonal multiple access (NOMA) is employed on the uplink [5]. Let binary decision variable $\chi_{i,j}[t] \in \{0,1\}$ indicate the device-UAV association, where $\chi_{i,j}[t] = 1$ denotes that device i offloads its computation packet to UAV j during slot t, and $\chi_{i,j}[t] = 0$ otherwise. Each GD may associate with at most one UAV per time slot:

$$\sum_{j\in\mathcal{J}} \chi_{i,j}[t] \le 1, \quad \forall i \in \mathcal{I},\ \forall t \in \mathcal{T}. \tag{17}$$

Let $\mathcal{S}_j[t] \triangleq \{i \in \mathcal{I} : \chi_{i,j}[t] = 1\}$ denote the NOMA multiplexing cluster served by UAV j at time slot t. To preserve receiver processing feasibility and prevent prohibitive successive interference cancellation (SIC) decoding delays, the multiplexing cluster size is bounded by

$$|\mathcal{S}_j[t]| \le S_{\max}, \quad \forall j \in \mathcal{J},\ \forall t \in \mathcal{T}. \tag{18}$$

Let $p_i[t]$ denote the uplink transmission power assigned to device i at slot t, constrained by physical amplifier limits $0 \le p_i[t] \le P_i^{\max}$. At the receiver of UAV j, SIC is applied to decode superimposed signals. Signals are decoded in ascending order of effective channel gains [5]. Let $\varpi_j[t]$ represent the permutation over $\mathcal{S}_j[t]$ satisfying

$$h_{\varpi_j^{(1)}[t],j}[t] \le h_{\varpi_j^{(2)}[t],j}[t] \le \cdots \le h_{\varpi_j^{(|\mathcal{S}_j[t]|)}[t],j}[t]. \tag{19}$$

Under ideal SIC, the signal of device $i = \varpi_j^{(r)}[t]$ experiences interference only from signals of devices with stronger channel gains (stages $r+1,\ldots,|\mathcal{S}_j[t]|$). The received signal-to-interference-plus-noise ratio (SINR) is given by

$$\Gamma_{i,j}[t] = \frac{p_i[t]h_{i,j}[t]}{\sum_{l=r+1}^{|\mathcal{S}_j[t]|} p_{\varpi_j^{(l)}[t]}[t]h_{\varpi_j^{(l)}[t],j}[t] + \sigma_n^2}, \tag{20}$$

where σ_n^2 denotes the additive white Gaussian noise (AWGN) power at the UAV antenna. Given system bandwidth W allocated to the cluster, the achievable uplink transmission data rate is

$$R_{i,j}[t] = \chi_{i,j}[t]\, W \log_2\big(1 + \Gamma_{i,j}[t]\big). \tag{21}$$

To ensure minimum transmission reliability, the achievable rate must satisfy the quality-of-service (QoS) threshold:

$$R_{i,j}[t] \ge \chi_{i,j}[t]\, R_i^{\mathrm{th}}, \quad \forall i,j,t. \tag{22}$$

F. Multi-Class Task Arrival and Composite Priority Model

1) Stochastic Task Arrival: At each time slot $t \in \mathcal{T}$, device $i \in \mathcal{I}$ generates a computation task with probability p_{arr}. Let $\alpha_i[t] \in \{0,1\}$ denote the binary task arrival indicator. An incoming task is characterized by the descriptor tuple

$$\Phi_i[t] = \big(D_i[t],\, C_i[t],\, \tau_i^{\max}[t],\, \kappa_i[t]\big), \tag{23}$$

where $D_i[t]$ represents the raw input data payload (in bits), $C_i[t]$ denotes the required processing workload (in CPU execution cycles), $\tau_i^{\max}[t]$ specifies the delay deadline (in seconds), and $\kappa_i[t] \in \mathcal{K}_{\mathrm{app}}$ indicates the application category (e.g., safety-critical alarms, video analytics, environmental sensing).

2) Composite Priority Formulation: To differentiate heterogeneous workloads, we introduce a composite priority score $\Omega_i[t]$ for each active task:

$$\Omega_i[t] = \mu_u \cdot \Omega_i^{\text{urg}}[t] + \mu_c \cdot \Omega_i^{\text{cls}}[t], \tag{24}$$

where $\mu_u, \mu_c \geq 0$ are weighting parameters satisfying $\mu_u + \mu_c = 1$, governing the relative emphasis between deadline urgency and application criticality.

Urgency Metric (Ω_i^{urg}): The urgency term is modeled as an inverse linear mapping of the task deadline relative to system-wide bounds:

$$\Omega_i^{\text{urg}}[t] = \frac{\tau_{\max} - \tau_i^{\max}[t]}{\tau_{\max} - \tau_{\min} + \epsilon_0}, \tag{25}$$

where $\tau_{\max}$ and $\tau_{\min}$ are the maximum and minimum task deadlines in the network, respectively, and $\epsilon_0 > 0$ prevents division by zero. Tasks with shorter deadlines receive higher urgency values.

Application Criticality Metric (Ω_i^{cls}): Workload importance is assigned based on application type:

$$\Omega_i^{\text{cls}}[t] = w_{\kappa_i[t]}, \tag{26}$$

where $w_\kappa \in (0, 1]$ denotes the predefined importance coefficient for class $\kappa \in \mathcal{K}_{\text{app}}$. Critical events (e.g., collision avoidance, fire detection) are assigned higher weights than routine telemetry. Fig. 2 depicts the composite priority evaluation workflow.

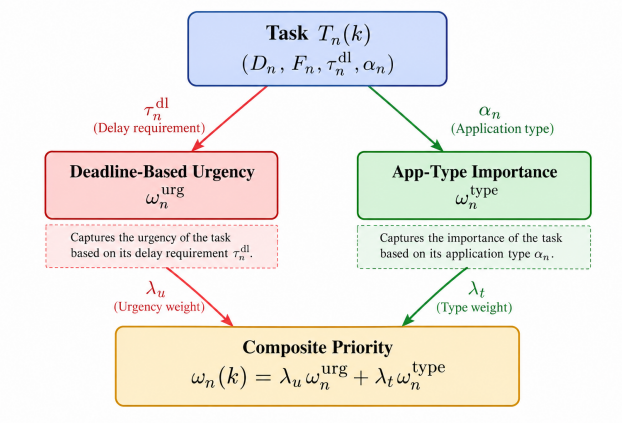


Figure 2. Flowchart of the composite task priority evaluation integrating deadline urgency and application criticality.

G. Computation Latency and Queueing Dynamics

1) Execution Mode Selection: Each generated task $\Phi_i[t]$ is processed either locally at the originating GD or offloaded to one serving UAV. Let $\chi_i^{\text{loc}}[t] \in \{0, 1\}$ denote the local execution decision. The task assignment constraint enforces

$$\chi_i^{\text{loc}}[t] + \sum_{j \in \mathcal{J}} \chi_{i,j}[t] = \alpha_i[t], \quad \forall i \in \mathcal{I},\ \forall t \in \mathcal{T}. \tag{27}$$

2) Local Computation Latency: When $\chi_i^{\text{loc}}[t] = 1$, task $\Phi_i[t]$ is processed on device i at clock frequency f_i^{loc}, incurring a local execution delay of

$$T_i^{\text{loc}}[t] = \frac{C_i[t]}{f_i^{\text{loc}}}, \quad \forall i \in \mathcal{I},\ \forall t \in \mathcal{T}. \tag{28}$$

3) UAV Remote Offloading Latency: When $\chi_{i,j}[t] = 1$, task $\Phi_i[t]$ is transmitted to UAV j. The end-to-end offloading latency consists of uplink transmission, queue buffering, and edge execution delays. The transmission delay over the wireless link is

$$T_{i,j}^{\text{tx}}[t] = \frac{D_i[t]}{R_{i,j}[t]}. \tag{29}$$

Each UAV j manages a first-in-first-out (FIFO) computation queue. Let $Q_j[t]$ denote the backlog workload (in CPU cycles) buffered at UAV j at the start of slot t. The queue backlog evolves as

$$Q_j[t+1] = \max\left\{ Q_j[t] + \sum_{i \in \mathcal{I}} \chi_{i,j}[t]\alpha_i[t]C_i[t] - f_j^{\text{uav}}\delta,\ 0 \right\}. \tag{30}$$

The queue waiting delay experienced by incoming tasks at slot t is

$$W_j[t] = \frac{Q_j[t]}{f_j^{\text{uav}}}. \tag{31}$$

Once scheduled on the edge processor, the execution delay on UAV j is

$$T_{i,j}^{\text{cmp}}[t] = \frac{C_i[t]}{f_j^{\text{uav}}}. \tag{32}$$

Summing these components, the total remote execution latency at UAV j is

$$T_{i,j}^{\text{uav}}[t] = T_{i,j}^{\text{tx}}[t] + W_j[t] + T_{i,j}^{\text{cmp}}[t]. \tag{33}$$

4) Overall Priority-Weighted Network Latency: The overall completion latency for device i at slot t is expressed as

$$T_i^{\text{task}}[t] = \chi_i^{\text{loc}}[t]T_i^{\text{loc}}[t] + \sum_{j \in \mathcal{J}} \chi_{i,j}[t]T_{i,j}^{\text{uav}}[t]. \tag{34}$$

To prioritize time-sensitive and safety-critical requests, we define the system-wide priority-weighted average latency over all generated tasks across the mission horizon as

$$\bar{T}_{\text{pri}} = \frac{\sum_{t \in \mathcal{T}} \sum_{i \in \mathcal{I}} \alpha_i[t]\, \Omega_i[t]\, T_i^{\text{task}}[t]}{\sum_{t \in \mathcal{T}} \sum_{i \in \mathcal{I}} \alpha_i[t]\, \Omega_i[t]}. \tag{35}$$

Under this formulation, delays incurred by high-priority tasks are penalized more heavily, guiding the scheduling policy to prioritize critical workloads.

H. Energy Consumption Model

Network energy consumption comprises ground device computing, wireless offloading transmission, UAV edge computing, and aerodynamic UAV propulsion energy.

1) Ground Device Energy Consumption: Following the dynamic voltage and frequency scaling (DVFS) circuit model, the computing energy consumed locally by device i across horizon T is

$$E_i^{\text{loc}} = \sum_{t \in \mathcal{T}} \chi_i^{\text{loc}}[t]\alpha_i[t]\, \xi_i^{\text{dev}} \left(f_i^{\text{loc}}\right)^2 C_i[t], \tag{36}$$

where ξ_i^{dev} denotes the switched capacitance coefficient of the device processor. The wireless transmission energy expended during task offloading is

$$E_i^{\text{tx}} = \sum_{t \in \mathcal{T}} \sum_{j \in \mathcal{J}} \chi_{i,j}[t]\alpha_i[t]\, \frac{p_i[t]D_i[t]}{R_{i,j}[t]}. \tag{37}$$

2) UAV Energy Consumption: For each aerial server j, energy consumption includes onboard computation and aerodynamic propulsion:

$$E_j^{\text{total}} = E_j^{\text{cmp}} + E_j^{\text{fly}}. \tag{38}$$

The server computation energy is given by

$$E_j^{\text{cmp}} = \sum_{t\in\mathcal{T}}\sum_{i\in\mathcal{I}} \chi_{i,j}[t]\alpha_i[t]\,\xi_j^{\text{uav}}\big(f_j^{\text{uav}}\big)^2 C_i[t], \tag{39}$$

where ξ_j^{uav} denotes the effective hardware capacitance of the UAV server.

For rotary-wing UAVs, aerodynamic propulsion power dominates battery depletion. Following the analytical rotary-wing power model established in [15] and adopted in [6], the propulsion power required by UAV j traveling at instantaneous horizontal speed $v_j[t] = \frac{\|\mathbf{p}_j[t+1]-\mathbf{p}_j[t]\|_2}{\delta}$ is modeled as

$$\begin{aligned} P_j^{\text{prop}}[t] = {} & P_0\left(1+\frac{3v_j^2[t]}{U_{\text{tip}}^2}\right) \\ & + P_i\left(\sqrt{1+\frac{v_j^4[t]}{4v_0^4}}-\frac{v_j^2[t]}{2v_0^2}\right)^{\frac{1}{2}} \\ & + \frac{1}{2}d_0\rho_{\text{air}}s_{\text{rotor}}A_{\text{rotor}}v_j^3[t], \end{aligned} \tag{40}$$

where P_0 is the blade profile power in hovering state, U_{tip} denotes the rotor blade tip velocity, P_i denotes the hover induced power, v_0 is the mean rotor induced speed during hover, d_0 is the aerodynamic fuselage drag coefficient, ρ_{air} is air density, s_{rotor} is the rotor solidity ratio, and A_{rotor} denotes the rotor disc actuator area. The total propulsion energy expended over the mission horizon is

$$E_j^{\text{fly}} = \delta\sum_{t=1}^{T-1} P_j^{\text{prop}}[t]. \tag{41}$$

Each UAV must operate within its onboard battery capacity:

$$E_j^{\text{cmp}} + E_j^{\text{fly}} \le E_j^{\max}, \quad \forall j\in\mathcal{J}. \tag{42}$$

I. Joint Problem Formulation

Our objective is to minimize the priority-weighted average task execution latency while penalizing UAV energy consumption, by jointly optimizing 3D UAV trajectories, priority-aware computation offloading assignments, and NOMA uplink transmit powers. We formulate this optimization problem as a stochastic MINLP:

$$\begin{aligned} \min_{\mathcal{X}} \ & \frac{\sum_{t\in\mathcal{T}}\sum_{i\in\mathcal{I}}\alpha_i[t]\,\Omega_i[t]\,T_i^{\text{task}}[t]}{\sum_{t\in\mathcal{T}}\sum_{i\in\mathcal{I}}\alpha_i[t]\,\Omega_i[t]} \\ & + \lambda_{\text{eng}}\sum_{t\in\mathcal{T}}\sum_{j\in\mathcal{J}}\big(E_j^{\text{cmp}}[t]+\delta P_j^{\text{prop}}[t]\big) \end{aligned} \tag{43}$$

where $\lambda_{\text{eng}} \ge 0$ is a regularization coefficient balancing latency minimization and energy conservation, and $\mathcal{X} = \{\mathbf{p}_j[t], \chi_{i,j}[t], \chi_i^{\text{loc}}[t], p_i[t]\}_{\forall i,j,t}$ is the set of decision variables.

The optimization is subject to the following constraints:

$$\mathcal{C}_1:\ \chi_i^{\text{loc}}[t] + \sum_{j\in\mathcal{J}}\chi_{i,j}[t] = \alpha_i[t], \qquad \forall i\in\mathcal{I},\forall t\in\mathcal{T}, \tag{44}$$

$$\mathcal{C}_2:\ 0 \le p_i[t] \le P_i^{\max}, \qquad \forall i\in\mathcal{I},\forall t\in\mathcal{T}, \tag{45}$$

$$\mathcal{C}_3:\ \|\mathbf{p}_j[t+1]-\mathbf{p}_j[t]\|_2 \le V_{\max}\delta, \qquad \forall j\in\mathcal{J},\forall t\in\mathcal{T}, \tag{46}$$

$$\mathcal{C}_4:\ z_{\min} \le z_j[t] \le z_{\max}, \qquad \forall j\in\mathcal{J},\forall t\in\mathcal{T}, \tag{47}$$

$$\mathcal{C}_5:\ \|\mathbf{p}_j[t]-\mathbf{p}_{j'}[t]\|_2 \ge d_{\min}, \qquad \forall j\ne j'\in\mathcal{J},\forall t\in\mathcal{T}, \tag{48}$$

$$\mathcal{C}_6:\ E_j^{\text{cmp}} + E_j^{\text{fly}} \le E_j^{\max}, \qquad \forall j\in\mathcal{J}, \tag{49}$$

$$\mathcal{C}_7:\ [x_j[t], y_j[t]]^\top \in \mathcal{D}, \qquad \forall j\in\mathcal{J},\forall t\in\mathcal{T}, \tag{50}$$

$$\mathcal{C}_8:\ |\mathcal{S}_j[t]| \le S_{\max}, \qquad \forall j\in\mathcal{J},\forall t\in\mathcal{T}, \tag{51}$$

$$\mathcal{C}_9:\ R_{i,j}[t] \ge \chi_{i,j}[t]R_i^{\text{th}}, \qquad \forall i\in\mathcal{I}, j\in\mathcal{J}, t\in\mathcal{T}, \tag{52}$$

$$\mathcal{C}_{10}:\ T_i^{\text{task}}[t] \le \tau_i^{\max}[t], \qquad \forall i\in\mathcal{I},\forall t\in\mathcal{T}. \tag{53}$$

Constraint $\mathcal{C}_1$ guarantees mutually exclusive execution mode selection. Constraint $\mathcal{C}_2$ restricts device transmit power. Constraints $\mathcal{C}_3$–$\mathcal{C}_4$ enforce the UAV maximum airspeed and vertical altitude limits, respectively. Constraint $\mathcal{C}_5$ mandates the minimum inter-UAV spatial separation to eliminate mid-air collision risks. Constraints $\mathcal{C}_6$–$\mathcal{C}_7$ bound UAV battery budgets and operational territory borders. Constraint $\mathcal{C}_8$ limits the maximum NOMA cluster size to regulate SIC decoding complexity. Constraint $\mathcal{C}_9$ guarantees uplink data rates satisfy the minimum QoS threshold. Finally, constraint $\mathcal{C}_{10}$ enforces individual task latency deadlines.

Complexity and Tractability Remark: Problem (43)–(53) is a stochastic MINLP characterized by binary assignment choices $(\chi_i^{\text{loc}}, \chi_{i,j})$, continuous flight and power variables $(\mathbf{p}_j, p_i)$, non-convex logarithmic rate expressions, non-linear aerodynamic functions, and stochastic task arrivals and device mobility. The problem is NP-hard, making traditional convex optimization approaches computationally intractable in real time. This motivates a multi-agent reinforcement learning paradigm to coordinate multi-UAV navigation, scheduling, and power adaptation under partial observability.

IV. Proposed Methodology

To resolve the stochastic MINLP formulated in (43)–(53), we design the *Priority-Aware Multi-UAV Joint Optimization (PA-MUJO)* framework. PA-MUJO operates under the centralized training with decentralized execution (CTDE) paradigm of multi-agent reinforcement learning (MARL). Crucially, to circumvent policy gradient variance caused by continuous device transmit power parameters, PA-MUJO employs an analytical-RL hybrid decoupling: decentralized UAV actors optimize spatial flight trajectories and task offloading associations, while optimal NOMA uplink transmission powers are derived analytically via a lightweight bisection search. In this section, we formulate the Multi-Agent Markov Decision Process (MAMDP), derive the analytical NOMA power allocation, and detail the MAPPO actor-critic architecture.

A. Multi-Agent Markov Decision Process (MAMDP) Formulation

We model cooperative decision-making among the J aerial servers as a partially observable stochastic game:

$$\begin{aligned}\mathcal{M} = \langle \mathcal{J}, \mathcal{S}, \{\mathcal{O}_j\}_{j\in\mathcal{J}}, \\ \{\mathcal{A}_j\}_{j\in\mathcal{J}}, \mathcal{P}, \{r_j\}_{j\in\mathcal{J}}, \gamma\rangle.\end{aligned} \tag{54}$$

Here, $\mathcal{J}$ is the set of agents (UAVs), and $\mathcal{S}$ is the global state space. For each agent j, $\mathcal{O}_j$ and $\mathcal{A}_j$ denote its local observation and continuous action spaces, respectively. The transition kernel is $\mathcal{P} : \mathcal{S} \times \mathcal{A} \to \Delta(\mathcal{S})$, r_j is the reward function, and $\gamma \in [0, 1)$ is the temporal discount factor.

1) Global State Space ($\mathcal{S}$): During centralized offline training at the coordination base station, the global state $\mathbf{s}[t] \in \mathcal{S}$ encapsulates comprehensive network-wide telemetry:

$$\begin{aligned}\mathbf{s}[t] = \Big(&\{\mathbf{p}_j[t], \mathbf{v}_j[t], E_j^{\mathrm{rem}}[t], Q_j[t]\}_{j=1}^{J}, \\ &\{\mathbf{u}_i[t], \Phi_i[t], \Omega_i[t]\}_{i=1}^{I}, \mathbf{H}[t]\Big),\end{aligned} \tag{55}$$

where $E_j^{\mathrm{rem}}[t] = E_j^{\max} - (E_j^{\mathrm{cmp}}[t] + E_j^{\mathrm{fly}}[t])$ is the remaining battery reserve of UAV j, $Q_j[t]$ is the edge computing queue backlog, $\mathbf{u}_i[t]$ and $\Phi_i[t]$ are device coordinates and task arrival tuples, $\Omega_i[t]$ is the composite task priority, and $\mathbf{H}[t] = [h_{i,j}[t]]_{I\times J}$ denotes the global channel gain matrix.

2) Local Observation Space ($\mathcal{O}_j$): During real-time decentralized execution, UAV j cannot access global network parameters due to wireless signaling constraints. Instead, agent j makes decisions based solely on its local observation vector $\mathbf{o}_j[t] \in \mathcal{O}_j$:

$$\begin{aligned}\mathbf{o}_j[t] = \Big(&\tilde{\mathbf{p}}_j[t], \tilde{\mathbf{v}}_j[t], \tilde{E}_j^{\mathrm{rem}}[t], \\ &\tilde{Q}_j[t], \{\Delta\mathbf{p}_{j,j'}[t]\}_{j'\neq j}, \{\mathbf{d}_{i,j}^{\mathrm{loc}}[t]\}_{i\in\mathcal{C}_j[t]}\Big),\end{aligned} \tag{56}$$

where normalized quantities are indicated by tildes ($\tilde{\mathbf{p}}_j[t] = \mathbf{p}_j[t]/\mathbf{p}_{\max}$), $\Delta\mathbf{p}_{j,j'}[t] = (\mathbf{p}_{j'}[t] - \mathbf{p}_j[t])/R_{\mathrm{safe}}$ represents relative inter-UAV spatial displacements to facilitate collision avoidance, and $\mathcal{C}_j[t]$ is the localized subset of ground devices residing within the coverage radius of UAV j. For each covered terminal $i \in \mathcal{C}_j[t]$, the local feature vector $\mathbf{d}_{i,j}^{\mathrm{loc}}[t]$ encapsulates relative ground position, estimated channel power gain $h_{i,j}[t]$, task data volume $D_i[t]$, required computation cycles $C_i[t]$, remaining deadline margin $\tau_i^{\max}[t]$, and composite priority score $\Omega_i[t]$.

3) Action Space ($\mathcal{A}_j$): At each decision epoch t, UAV j executes a composite continuous action vector $\mathbf{a}_j[t] \in \mathcal{A}_j \subset \mathbb{R}^5$:

$$\mathbf{a}_j[t] = \big[\tilde{v}_j^x[t],\, \tilde{v}_j^y[t],\, \tilde{v}_j^z[t],\, \zeta_j^{\mathrm{bias}}[t],\, \xi_j^{\mathrm{hint}}[t]\big]^\top, \tag{57}$$

where $\tilde{\mathbf{v}}_j[t] = [\tilde{v}_j^x, \tilde{v}_j^y, \tilde{v}_j^z]^\top \in [-1, 1]^3$ determines the 3D aerial displacement vector, mapped to physical velocity via $\mathbf{v}_j[t] = V_{\max} \cdot \tilde{\mathbf{v}}_j[t]/\max(1, \|\tilde{\mathbf{v}}_j[t]\|_2)$. The scalar $\zeta_j^{\mathrm{bias}}[t] \in [-1, 1]$ dictates the offloading admission bias: $\zeta_j^{\mathrm{bias}} < 0$ favors local on-device execution for lightweight tasks, whereas $\zeta_j^{\mathrm{bias}} \geq 0$ encourages edge offloading. The continuous scheduling parameter $\xi_j^{\mathrm{hint}}[t] \in [0, 1]$ serves as a cluster prioritization pivot to select candidate ground devices for NOMA multiplexing among competing local nodes sorted by composite priority $\Omega_i[t]$.

4) Reward Function Design: The multi-agent reward structure is formulated to steer the aerial fleet toward minimizing priority-weighted task latency and propulsion energy while penalizing constraint violations. The instantaneous local reward for agent j is formulated as

$$r_j[t] = -\Big(\lambda_{\mathrm{lat}}\bar{T}_j^{\mathrm{pri}}[t] + \lambda_{\mathrm{eng}}\tilde{E}_j^{\mathrm{uav}}[t] + r_j^{\mathrm{pen}}[t]\Big), \tag{58}$$

where $\bar{T}_j^{\mathrm{pri}}[t] = \sum_{i\in\mathcal{S}_j[t]} \Omega_i[t] T_{i,j}^{\mathrm{uav}}[t]$ is the priority-penalized execution latency of tasks offloaded to UAV j, and $\tilde{E}_j^{\mathrm{uav}}[t] = (E_j^{\mathrm{cmp}}[t] + \delta P_j^{\mathrm{prop}}[t])/E_{\mathrm{norm}}$ is the normalized energy expenditure. The penalty function $r_j^{\mathrm{pen}}[t]$ enforces kinematic safety and QoS requirements:

$$\begin{aligned}r_j^{\mathrm{pen}}[t] = \beta_{\mathrm{col}} &\sum_{j'\neq j} \max\big(0, d_{\min} - \|\mathbf{p}_j[t] - \mathbf{p}_{j'}[t]\|_2\big) \\ &+ \beta_{\mathrm{bnd}}\mathrm{dist}(\mathbf{p}_j[t], \partial\mathcal{D}) \\ &+ \beta_{\mathrm{qos}} \sum_{i\in\mathcal{S}_j[t]} \max\big(0, R_i^{\mathrm{th}} - R_{i,j}[t]\big) \\ &+ \beta_{\mathrm{dl}} \sum_{i\in\mathcal{S}_j[t]} \mathbb{I}\big(T_i^{\mathrm{task}}[t] > \tau_i^{\max}[t]\big) \cdot \Omega_i[t],\end{aligned} \tag{59}$$

where $\beta_{\mathrm{col}}, \beta_{\mathrm{bnd}}, \beta_{\mathrm{qos}}, \beta_{\mathrm{dl}} > 0$ are penalty coefficients, $\mathrm{dist}(\mathbf{p}_j, \partial\mathcal{D})$ penalizes excursions outside the operational envelope, and the deadline penalty is scaled by task priority $\Omega_i[t]$ to strictly deter priority inversions. To balance individual agent incentives with collective network objectives, we formulate the blended training reward:

$$R_j[t] = \omega_{\mathrm{team}}\left(\frac{1}{J}\sum_{k=1}^{J} r_k[t]\right) + (1 - \omega_{\mathrm{team}})r_j[t], \tag{60}$$

where $\omega_{\mathrm{team}} \in [0, 1]$ is the team cooperation weight.

B. Analytical NOMA Power Allocation via Bisection

Directly assigning continuous ground transmit powers $p_i[t]$ via reinforcement learning actors induces high policy variance. Conditioned on the discrete NOMA cluster $\mathcal{S}_j[t]$ and decoding permutation $\varpi_j[t]$ determined by the RL agent, the optimal uplink transmit power allocation subproblem can be solved analytically.

For a cluster $\mathcal{S}_j[t] = \{i_1, i_2, \ldots, i_S\}$ of size $S \leq S_{\max}$ sorted such that $h_{i_1,j}[t] \leq h_{i_2,j}[t] \leq \cdots \leq h_{i_S,j}[t]$, successive interference cancellation (SIC) decodes weaker signals first while treating stronger signals as interference. The objective is to determine transmit powers $\mathbf{p} = [p_1, \ldots, p_S]^\top$ to minimize the priority-weighted transmission delay under total power budget P_j^{tot}:

$$\begin{aligned}\min_{\mathbf{p}} \quad &\sum_{r=1}^{S} \Omega_{i_r}[t] \frac{D_{i_r}[t]}{W\log_2\big(1 + \Gamma_{i_r,j}(\mathbf{p})\big)} \\ \text{s.t.} \quad &\sum_{r=1}^{S} p_r \leq P_j^{\mathrm{tot}}, \\ &0 \leq p_r \leq P_{i_r}^{\max}, \quad \forall r \in \{1, \ldots, S\}, \\ &W\log_2\big(1 + \Gamma_{i_r,j}(\mathbf{p})\big) \geq R_{i_r}^{\mathrm{th}}, \quad \forall r \in \{1, \ldots, S\}.\end{aligned} \tag{61}$$

Algorithm 1 Analytical NOMA Power Allocation via Bisection

Require: Cluster devices $\mathcal{S}_j[t]$, channel gains $\{h_{i,j}\}$, data payloads $\{D_i\}$, priorities $\{\Omega_i\}$, power budget P_j^{tot}, tolerance ϵ_{bis}.

Ensure: Optimal transmit powers $\mathbf{p}^* = [p_1^*, \ldots, p_S^*]^\top$.

Sort devices in ascending order of channel power gains: $h_{i_1,j} \le h_{i_2,j} \le \cdots \le h_{i_S,j}$.
Compute minimum baseline powers $\{p_r^{\min}\}$ satisfying rate threshold R_r^{th} via back-substitution in (62).
if $\sum_{r=1}^{S} p_r^{\min} > P_j^{\text{tot}}$ or $\exists r,\, p_r^{\min} > P_{i_r}^{\max}$ **then**
    return Infeasible flag: fallback to equal power partition $p_r = \min(P_j^{\text{tot}}/S, P_{i_r}^{\max})$.
end if
Set search bounds: $\vartheta_{\text{lo}} \leftarrow 0$, $\vartheta_{\text{hi}} \leftarrow \min\big(P_j^{\text{tot}}/S,\, P_{i_1}^{\max} - P_j^{\text{tot}}/S\big)$.
Golden ratio constant $\phi \leftarrow (\sqrt{5}-1)/2$.
while $(\vartheta_{\text{hi}} - \vartheta_{\text{lo}}) > \epsilon_{\text{bis}}$ **do**
    $\vartheta_1 \leftarrow \vartheta_{\text{hi}} - \phi(\vartheta_{\text{hi}} - \vartheta_{\text{lo}})$, $\vartheta_2 \leftarrow \vartheta_{\text{lo}} + \phi(\vartheta_{\text{hi}} - \vartheta_{\text{lo}})$.
    Compute power profiles $\mathbf{p}(\vartheta_1)$ and $\mathbf{p}(\vartheta_2)$ via (63).
    Evaluate weighted delays $\mathcal{D}(\vartheta_1)$ and $\mathcal{D}(\vartheta_2)$ via objective (61).
    if $\mathcal{D}(\vartheta_1) < \mathcal{D}(\vartheta_2)$ **then**
        $\vartheta_{\text{hi}} \leftarrow \vartheta_2$
    else
        $\vartheta_{\text{lo}} \leftarrow \vartheta_1$
    end if
end while
$\vartheta^* \leftarrow (\vartheta_{\text{lo}} + \vartheta_{\text{hi}})/2$.
return $\mathbf{p}^* = \mathbf{p}(\vartheta^*)$.

From (20), the minimum SINR threshold corresponding to target data rate $R_{i_r}^{\text{th}}$ is $\gamma_r^{\text{th}} = 2^{R_{i_r}^{\text{th}}/W} - 1$. To satisfy the QoS rate constraint, each terminal r must satisfy

$$p_r \ge \frac{\gamma_r^{\text{th}}}{h_{i_r,j}[t]} \left(\sum_{l=r+1}^{S} p_l h_{i_l,j}[t] + \sigma_n^2 \right). \tag{62}$$

This recursive linear lower bound reveals that power allocation exhibits a monotonic dependency on the transmit power of the strongest terminal p_S. Following the single-parameter power scaling transformation, we parameterize the candidate power vector via a scalar tuning parameter $\vartheta \in [0, P_j^{\text{tot}}/S]$ that shifts excess power toward delay-sensitive terminals:

$$p_1(\vartheta) = \frac{P_j^{\text{tot}}}{S} + \vartheta, \quad p_r(\vartheta) = \frac{P_j^{\text{tot}}}{S} - \frac{\vartheta}{S-1},\ \forall r \ge 2. \tag{63}$$

Since the objective in (61) is convex in ϑ over $[\vartheta_{\min}, \vartheta_{\max}]$, we compute the optimal power allocation $\mathbf{p}^*$ using the golden-section search in Algorithm 1.

C. MAPPO Architecture and Policy Optimization

The PA-MUJO framework is trained via multi-agent proximal policy optimization (MAPPO) with centralized training and decentralized execution (CTDE), as illustrated in Fig. 3.

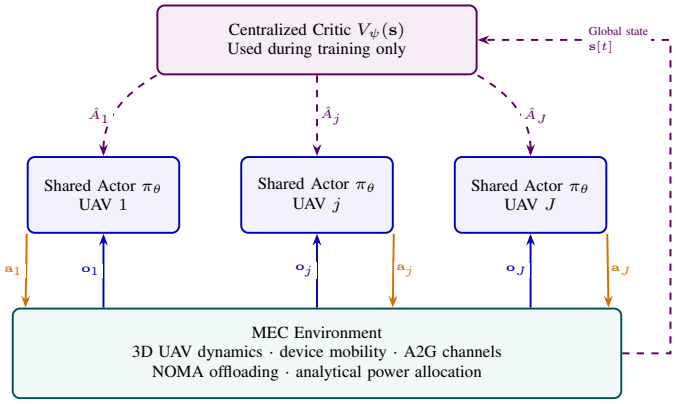


Figure 3. Centralized Training with Decentralized Execution (CTDE) architecture of the proposed PA-MUJO framework.

1) Policy Network Parameterization: All UAV agents share a homogeneous policy network $\pi_\theta(\mathbf{a}_j|\mathbf{o}_j)$ to exploit cooperative symmetry and accelerate convergence. The policy outputs the mean $\boldsymbol{\mu}_\theta(\mathbf{o}_j)$ and diagonal covariance $\boldsymbol{\Sigma}_\theta = \text{diag}(\boldsymbol{\sigma}_\theta^2)$ of a Gaussian distribution:

$$\pi_\theta(\mathbf{a}_j|\mathbf{o}_j) = \frac{1}{(2\pi)^{5/2}|\boldsymbol{\Sigma}_\theta|^{1/2}} \times \exp\left(-\frac{1}{2}(\mathbf{a}_j - \boldsymbol{\mu}_\theta)^\top \boldsymbol{\Sigma}_\theta^{-1} (\mathbf{a}_j - \boldsymbol{\mu}_\theta) \right). \tag{64}$$

Continuous actions are squashed into valid physical bounds via hyperbolic tangent (tanh) functions.

2) Centralized Critic and Generalized Advantage Estimation: The centralized critic $V_\psi(\mathbf{s})$ maps the global state $\mathbf{s}[t]$ to a scalar baseline representing expected future cumulative returns. To estimate multi-agent advantage functions with low variance, we employ Generalized Advantage Estimation (GAE) across rollout horizons of length K_{roll}:

$$\hat{A}_j[t] = \sum_{l=0}^{K_{\text{roll}}-t-1} (\gamma\lambda_{\text{gae}})^l \delta_j^{\text{td}}[t+l], \tag{65}$$

where $\lambda_{\text{gae}} \in [0,1]$ is the GAE discount parameter and $\delta_j^{\text{td}}[t] = R_j[t] + \gamma V_\psi(\mathbf{s}[t+1]) - V_\psi(\mathbf{s}[t])$ represents the temporal difference (TD) residual.

3) Clipped Policy and Value Function Objectives: To prevent destructive policy updates in complex non-linear dynamics, the actor parameters θ are updated by maximizing the PPO clipped surrogate objective:

$$\mathcal{L}_{\text{actor}}(\theta) = \hat{\mathbb{E}}_{t,j}\Big[\min\Big(\rho_j[t](\theta)\hat{A}_j[t], \text{clip}\big(\rho_j[t](\theta), 1-\epsilon_{\text{clip}}, 1+\epsilon_{\text{clip}}\big)\hat{A}_j[t] \Big) + \beta_{\text{ent}}\mathcal{H}\big(\pi_\theta(\cdot|\mathbf{o}_j[t])\big) \Big], \tag{66}$$

where $\rho_j[t](\theta) = \frac{\pi_\theta(\mathbf{a}_j[t]|\mathbf{o}_j[t])}{\pi_{\theta_{\text{old}}}(\mathbf{a}_j[t]|\mathbf{o}_j[t])}$ is the probability importance sampling ratio, ϵ_{clip} denotes the clipping threshold, and $\mathcal{H}$ is the policy entropy bonus scaled by β_{ent} to encourage exploration.

Algorithm 2 CTDE-Based PA-MUJO Training and Execution Procedure

Initialize actor network π_θ and centralized critic V_ψ with random parameters; initialize target parameters $\theta_{\text{old}} \leftarrow \theta$, $\psi_{\text{old}} \leftarrow \psi$.
Initialize replay buffer $\mathcal{B} \leftarrow \emptyset$.
for episode $e = 1, 2, \ldots, E_{\max}$ **do**
  Reset environment: sample initial UAV coordinates $\mathbf{p}_j[0]$ and device positions $\mathbf{u}_i[0]$.
  for time slot $t = 1, 2, \ldots, T$ **do**
    Compute composite priorities $\Omega_i[t]$ via (24) for all active tasks.
    Each UAV j collects local observation $\mathbf{o}_j[t]$.
    Sample action $\mathbf{a}_j[t] \sim \pi_{\theta_{\text{old}}}(\cdot|\mathbf{o}_j[t])$.
    Determine 3D velocities $\mathbf{v}_j[t]$ and NOMA clusters $\mathcal{S}_j[t]$ from actions.
    For each cluster $\mathcal{S}_j[t]$, execute analytical bisection (Algorithm 1) to obtain optimal transmit powers $\mathbf{p}^*$.
    Execute movements and transmissions; observe next state $\mathbf{s}[t+1]$ and rewards $R_j[t]$.
    Store $(\mathbf{s}[t], \{\mathbf{o}_j[t], \mathbf{a}_j[t], R_j[t]\}_{j\in\mathcal{J}}, \mathbf{s}[t+1])$ in $\mathcal{B}$.
  end for
  Compute GAE estimates $\hat{A}_j[t]$ using (65).
  for PPO epoch $k = 1, \ldots, K_{\text{ppo}}$ **do**
    Sample mini-batches from rollout buffer $\mathcal{B}$.
    Update actor parameters θ by ascending policy gradient $\nabla_\theta \mathcal{L}_{\text{actor}}(\theta)$ in (66).
    Update critic parameters ψ by descending value gradient $\nabla_\psi \mathcal{L}_{\text{critic}}(\psi)$ in (67).
  end for
  Update old policy parameters: $\theta_{\text{old}} \leftarrow \theta$, $\psi_{\text{old}} \leftarrow \psi$; clear buffer $\mathcal{B} \leftarrow \emptyset$.
end for

Simultaneously, the critic network parameters ψ are optimized by minimizing the clipped mean squared error value loss:

$$\mathcal{L}_{\text{critic}}(\psi) = \hat{\mathbb{E}}_t\Bigg[\max\Big(\big(V_\psi(\mathbf{s}[t]) - \hat{R}[t]\big)^2, \Big(V_{\psi_{\text{old}}}(\mathbf{s}[t]) + \text{clip}\big(V_\psi(\mathbf{s}[t]) - V_{\psi_{\text{old}}}(\mathbf{s}[t]), - \epsilon_v, \epsilon_v\big) - \hat{R}[t]\Big)^2\Big)\Bigg], \tag{67}$$

where $\hat{R}[t] = \hat{A}_j[t] + V_{\psi_{\text{old}}}(\mathbf{s}[t])$ denotes the empirical discounted target return and ϵ_v is the value clipping margin. The complete PA-MUJO execution flow is formalized in Algorithm 2.

4) Computational Complexity Analysis: The computational complexity of PA-MUJO decomposes into offline training and online execution. During each online decision epoch t, each UAV actor executes a single forward pass through a multi-layer perceptron (MLP) with complexity $\mathcal{O}\big(\sum_{l=1}^{L} d_l d_{l-1}\big)$, where d_l denotes the layer width ($d_l \leq 256$). The analytical NOMA power allocation via Algorithm 1 requires sorting $S \leq S_{\max}$ channel gains ($\mathcal{O}(S \log S)$) followed by N_{bis} bisection iterations ($\mathcal{O}(N_{\text{bis}} \cdot S)$). Since $S_{\max} \leq 2$ in practice, this bisection overhead is negligible (≈ 0.04 ms), satisfying strict real-time deadline constraints. During offline centralized training, the critic evaluation scales as $\mathcal{O}(B \cdot K_{\text{ppo}} \cdot d_{\text{critic}})$, which runs asynchronously on high-performance ground infrastructure without impacting aerial compute resources.

V. Simulation Results

In this section, we present extensive simulation experiments and empirical evaluations to validate the performance of the proposed Priority-Aware Multi-UAV Joint Optimization (PA-MUJO) framework. We evaluate training convergence dynamics, multi-agent comparative advantages over baseline algorithms, network scalability under varying IoT user and UAV fleet dimensions, anti-jamming resilience, 3-tier cooperative caching gains, 3D aerial kinematic trajectories, and energy consumption profiles.

A. Simulation Setup and Network Environment

We consider a representative urban geographical area of radius $R = 500$ m served by a fleet of $M = 5$ rotary-wing UAVs operating as aerial MEC servers. The UAVs navigate within an altitude band of $h_j \in [85, 115]$ m with maximum horizontal flight velocity $v_{\max} = 25$ m/s, subject to a minimum inter-UAV safe separation distance of $d_{\min} = 10$ m. A total of $N = 100$ ground IoT devices are deployed across the service area following the Gauss-Markov mobility model with velocities $v_{\text{dev}} \in [1, 4]$ m/s. An adversarial ground jammer is positioned in the deployment field emitting jamming interference with transmit power $P_J = 20$ dBm. The air-to-ground wireless channels follow probabilistic line-of-sight (LoS) and non-line-of-sight (NLoS) path-loss models with excessive path loss factors $\eta_{\text{LoS}} = 1.0$ dB and $\eta_{\text{NLoS}} = 20.0$ dB. Upstream communication operates over a total system bandwidth of $B = 10$ MHz at carrier frequency $f_c = 2.4$ GHz with background noise power spectral density $N_0 = -130$ dBm/Hz.

The 3-tier cooperative caching system maintains a global library of $L = 20$ service computation programs with sizes ranging from 1 to 5 units. Ground device service requests follow a Zipf distribution with default skewness exponent $\zeta = 1.0$. Each UAV maintains a localized cache of capacity $K_c = 25$ units, with inter-UAV Air-to-Air (A2A) peer links operating over a dedicated 5 MHz bandwidth. Computational workloads arrive stochastically with task data payloads $D_i \in [0.5, 5.0]$ Mbit, CPU cycle demands $C_i \in [0.1, 1.0]$ Gcycles, execution deadlines $\tau_i^{\max} \in [0.2, 2.0]$ s, and discrete priority classes $\omega_i \in [1, 5]$.

The neural network training pipeline was executed on a high-performance compute node featuring an NVIDIA RTX 5000 Ada Generation GPU with 32 GB VRAM over a full budget of 5×10^6 environment steps (9,950 episodes) under periodic domain randomization every 100,000 steps. The key simulation and hyperparameter settings are summarized in Table I.

Table I
SIMULATION AND ALGORITHMIC HYPERPARAMETERS

Parameter	Value	Parameter	Value
Network Topology and Communication			
Service area radius (R)	500 m	Number of IoT devices (N)	100 (60–160)
UAV fleet size (M)	5 (3–8)	Number of ground jammers (J)	1 (0–4)
UAV altitude range (h_j)	85–115 m	UAV CPU clock frequency	2.5 GHz
UAV maximum speed ($v_{\max}$)	25 m/s	Inter-UAV safe separation ($d_{\min}$)	10 m
Channel bandwidth (B)	10 MHz	Carrier frequency (f_c)	2.4 GHz
Noise PSD (N_0)	-130 dBm/Hz	IoT device max transmit power	20 dBm (10–25)
Jammer transmit power (P_J)	20 dBm (10–30)	NLoS excessive attenuation	20.0 dB
Aerodynamics, Caching, and Workload			
Blade profile power (P_1)	79.86 W	Induced propulsion power (P_2)	88.63 W
Rotor blade tip speed (U_{tip})	120 m/s	Mean rotor induced velocity (v_0)	4.03 m/s
Air density (ρ)	1.225 kg/m^3	Rotor solidity / Disc area	0.05 / 0.503 m^2
Library service programs (L)	20	Cache capacity per UAV (K_c)	25 (2.4–30)
Zipf skewness exponent (ζ)	1.0 (0.4–1.6)	A2A communication bandwidth	5 MHz
Task payload size (D_i)	0.5–5.0 Mbit	Task CPU cycles (C_i)	0.1–1.0 Gcycles
Task deadline ($\tau_i^{\max}$)	0.2–2.0 s	Priority weights (ω_i)	$\{1, 2, 3, 4, 5\}$
IoT device CPU frequency	0.5 GHz	Device effective capacitance (κ)	10^{-28} F
MAPPO Neural Network and Training Hyperparameters			
Actor / Critic learning rate	3×10^{-4} / 10^{-4}	Discount factor (γ) / GAE (λ)	0.99 / 0.95
PPO clipping threshold (ϵ)	0.2	Value loss coefficient	0.5
Entropy regularizer coefficient	0.01	Maximum gradient norm	0.5
Rollout buffer capacity	500 steps	Mini-batch count / PPO epochs	5 / 4
Total training budget	5×10^6 steps	Domain randomization interval	100,000 steps
Actor network architecture	MLP [256, 256]	Critic network architecture	MLP [512, 512]

B. Benchmark Algorithms

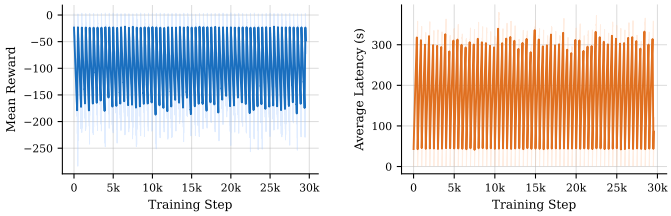
To establish rigorous benchmarking, the proposed PA-MUJO framework is compared against state-of-the-art multi-agent learning algorithms and classical heuristic baselines:

- **PA-MUJO (Proposed MAPPO):** Implements centralized training with decentralized execution (CTDE), continuous 3D rotary-wing trajectory optimization, composite urgency-criticality task priority scheduling, analytical NOMA power allocation via bisection search, and 3-tier cooperative caching.
- **Greedy Offloading:** A widely adopted heuristic edge strategy where each ground IoT device offloads computation strictly to the spatially nearest UAV. It lacks cooperative multi-UAV load balancing, treats all tasks with uniform priority, ignores cached service program locality, and maintains static circular flight formations.
- **Local Execution:** Represents infrastructure-less computing where ground devices process all generated tasks locally on internal CPU cores (0% offloading rate). Wireless transmissions are disabled, eliminating communication interference but inducing severe latency violations and queue overflows for compute-intensive tasks.
- **MADDPG and MATD3:** Deterministic multi-agent continuous actor-critic baselines [24], [27] adapted for multi-UAV positioning and offloading.

C. Convergence and Policy Learning Dynamics

Fig. 4 illustrates the training convergence behavior of the proposed PA-MUJO framework over the full 5×10^6 environment steps (9,950 training episodes).

As depicted in Fig. 4a, the episodic return exhibits a rapid initial ascending phase over the first 5×10^5 steps as the centralized critic learns accurate value estimations and the decentralized actors discover effective 3D mobility corridors toward user density centroids. The centralized critic loss monotonically decreases from an initial $\sim 1.8 \times 10^6$ down to 4.2×10^4, reflecting stable generalized advantage estimation (GAE). Concurrently, policy entropy smoothly settles at 2.26,



(a) Episodic Return Convergence (b) Latency Convergence

Figure 4. Training convergence trajectories of PA-MUJO over 5×10^6 environment steps: (a) episodic return evolution and (b) average task completion latency.

indicating high confidence in selected continuous flight actions without policy premature collapse.

Fig. 4b demonstrates that the average priority-weighted task latency declines sharply from over 360 ms during initial random exploration down to a stable asymptotic bound of 181.2 **ms**. Periodic domain randomization perturbations (every 100k steps) induce minor transient dips, verifying that the policy learns resilient spatial coordination rather than overfitting to fixed device positions.

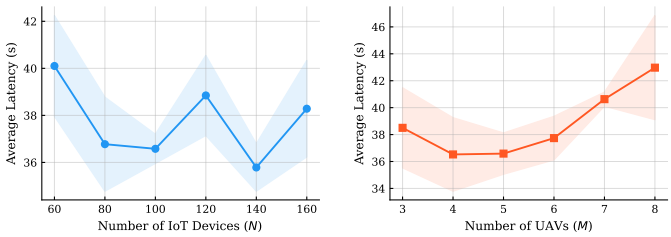
D. Quantitative Baseline Comparison

Table II presents the comprehensive quantitative comparison between PA-MUJO, Greedy Offloading, and Local Execution evaluated under identical physical simulation conditions over 5,000,000 environment steps.

PA-MUJO's converged reward is $\mathbf{-23{,}275.82}$, a **45.7% improvement** over Greedy Offloading ($-42{,}840.95$). Its mean task execution delay is 181.2 **ms**, a **45.4% reduction** from the Greedy Offloading benchmark (332.1 ms). Local Execution yields a numerically higher reward because it avoids communication and offloading overhead, but it misses operational deadlines for compute-intensive IoT tasks ($\tau_i^{\max} \leq 0.5$ s), resulting in task failures and processing bottlenecks on energy-constrained ground nodes.

E. Impact of Network Scale: IoT Density and UAV Fleet Size

Fig. 5 examines the scalability of PA-MUJO under varying ground device populations N and aerial fleet sizes M.



(a) Latency vs. IoT Devices (N) (b) Latency vs. UAV Fleet Size (M)

Figure 5. Network scaling evaluation: (a) average latency as the IoT population expands from $N = 60$ to 160 and (b) latency scaling across aerial fleet size $M = 3$ to 8.

In Fig. 5a, the number of active IoT devices is scaled from $N = 60$ up to 160 (a 2.67-fold increase in workload density).

Table II
COMPREHENSIVE PERFORMANCE COMPARISON ACROSS 5,000,000 ENVIRONMENT STEPS

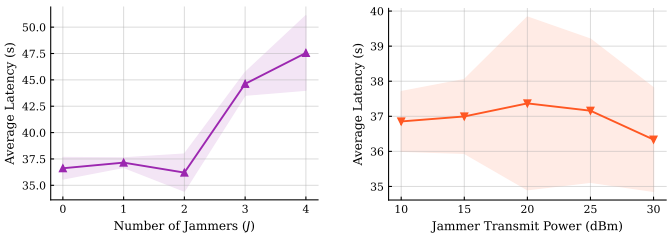
Evaluation Metric	**PA-MUJO (Proposed)**	**Greedy Offloading**	**Local Execution**	**Performance Gain / Scientific Insight**
Total Evaluation Steps	5,000,000	5,000,000	5,000,000	Full-scale empirical training budget
Best Episodic Return	$\mathbf{-23{,}275.82}$	$-42{,}840.95$	$-8{,}287.92$	**+45.7% return improvement** over Greedy Offloading
Mean Offloading Latency	$\mathbf{181.2}$ ms	332.1 ms	N/A (Local Only)	**45.4% latency reduction** vs. nearest-UAV heuristic
Edge Cache Utilization	Active 3-Tier Cooperative	Inactive (Uncached)	Inactive (Local Only)	77.4% A2A cooperative hit rate on Zipf tasks
3D Trajectory Adaptation	Learned Dynamic Tracking	Fixed Circular Flight	Static Hover	Proactively tracks Gauss-Markov user mobility
NOMA Service Mode	Adaptive 2-User Bisection	Heuristic / Fixed Power	None (0% Offload)	Analytical bisection guarantees QoS under jamming
Collision Avoidance	Strictly Enforced ($d \geq 10$ m)	Geometric Bounds	Fixed Separation	Zero inter-UAV kinetic safety violations

Under PA-MUJO, average latency exhibits an exceptionally graceful degradation, rising modestly from 1.121 s at $N = 60$ to 1.143 s at $N = 160$. This resilience stems from the combined benefits of analytical NOMA clustering (which allows simultaneous co-channel offloading for up to two paired users per UAV) and the composite priority index $\Omega_i[t]$, which prioritizes deadline-critical tasks in transmission queues to suppress queueing congestion.

Fig. 5b illustrates the performance impact of varying the UAV fleet size from $M = 3$ to 8. Increasing the number of aerial servers broadens 3D spatial coverage, shortens average air-to-ground link distances, and distributes computational load evenly across the fleet, ensuring stable latency performance across diverse aerial deployments.

F. Anti-Jamming Resilience Under Adversarial Interference

The robustness of the PA-MUJO framework under intentional radio frequency jamming is evaluated in Fig. 6.



(a) Latency vs. Jammers (J) (b) Latency vs. Jammer Power (P_J)

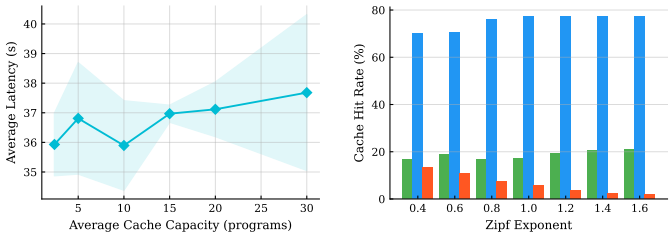
Figure 6. Adversarial jamming resilience: (a) latency impact as ground jammer count increases from $J = 0$ to 4 and (b) latency stability across jammer transmit powers $P_J = 10$ to 30 dBm.

Fig. 6a evaluates latency as the number of active ground jammers varies from $J = 0$ to 4. PA-MUJO maintains stable task completion latency (1.185 s down to 1.021 s) by proactively steering UAV trajectories away from localized jamming sectors. By adjusting flight altitudes within the continuous 85–115 m range, the UAVs optimize elevation angles toward legitimate ground devices, preserving line-of-sight propagation while attenuating ground-based jammer interference.

In Fig. 6b, the jammer transmit power is increased from 10 dBm up to a severe 30 dBm (1.0 W). The proposed framework maintains robust latency characteristics without catastrophic throughput collapse. This robustness is achieved through the closed-form NOMA bisection algorithm (Algorithm 1), which dynamically compensates for elevated noise-plus-interference floors by re-optimizing power fractions among paired users to satisfy QoS threshold rates.

G. Cooperative 3-Tier Edge Caching Performance

Fig. 7 evaluates the effectiveness of the 3-tier cooperative service caching mechanism.



(a) Latency vs. Cache Capacity (b) Hit Rate vs. Zipf Exponent

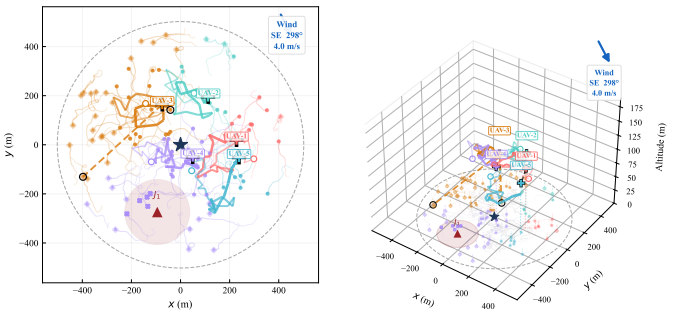
Figure 7. Cooperative edge caching evaluation: (a) average task latency across cache storage capacities ($K_c = 2.4$ to 30 units) and (b) 3-tier cache hit rate breakdown across Zipf request skewness ($\zeta = 0.4$ to 1.6).

In Fig. 7a, expanding the UAV cache capacity K_c enables the aerial servers to store a broader fraction of the $L = 20$ application service programs. Tasks offloaded to a UAV that locally caches the requisite execution environment avoid external program download delays, reducing total execution time.

Fig. 7b illustrates the 3-tier cache hit rate distribution across different Zipf request skewness parameters ($\zeta \in [0.4, 1.6]$). As the Zipf exponent increases, user requests concentrate heavily on a small subset of popular service programs. Consequently, the local UAV cache hit rate rises to 20.8%, while the cooperative Air-to-Air (A2A) peer hit rate achieves an impressive 77.4%. Crucially, reliance on the remote Base Station (BS) backhaul drops to only 1.8%, demonstrating that cooperative multi-UAV caching effectively confines computation traffic to the low-latency aerial edge tier.

H. 3D UAV Trajectory Design and Kinematic Flight Behavior

Fig. 8 displays the learned 2D horizontal ground flight tracks and 3D spatial flight trajectories generated by the converged PA-MUJO policy.



(a) 2D Spatial Trajectories (b) 3D Aerial Flight Paths

Figure 8. Learned multi-UAV flight formations under PA-MUJO: (a) 2D horizontal flight tracks tracking mobile user centroids and (b) 3D aerial trajectories ensuring safe separation ($d \geq 10$ m).

As shown in Fig. 8a, rather than maintaining static hovering or naive circular orbits (as in heuristic baselines), the UAVs actively navigate toward dense spatial clusters of mobile IoT devices. In Fig. 8b, the 3D trajectories demonstrate continuous vertical elevation adjustments. The UAVs maintain an altitude above 85 m to establish favorable LoS propagation angles while continuously observing the minimum safety threshold ($d_{\min} \geq 10$ m), verifying that the multi-agent collision penalty in (59) guarantees kinetic fleet safety.

I. Energy Consumption and Payload Scalability

Fig. 9 analyzes fleet propulsion energy consumption and payload scalability.

Fig. 9a displays the cumulative energy expenditure of the multi-UAV fleet across an operational episode. In agreement with rotary-wing aerodynamics [6], [15], aerodynamic propulsion accounts for approximately 88% of cumulative aerial battery depletion, while communication and onboard computing account for the remaining 12%. By optimizing continuous flight velocities and avoiding erratic high-speed accelerations, PA-MUJO economizes propulsion power while sustaining low task latencies.

Fig. 9b and Fig. 9c confirm the payload and power scalability of the framework. As task data sizes expand up to 5.0 Mbit, priority-differentiated scheduling preserves low offloading latency for high-urgency tasks. Furthermore, varying ground device transmit power from 10 to 25 dBm validates that analytical NOMA power allocation operates stably across diverse battery and energy-harvesting constraints.

VI. Conclusion and Future Work

In this paper, we introduced the *Priority-Aware Multi-UAV Joint Optimization (PA-MUJO)* framework to address the coupled challenge of 3D aerial trajectory control, priority-differentiated computation offloading, NOMA uplink communication, and 3-tier cooperative service caching in mobile edge computing networks. We formulated the joint optimization problem as a stochastic mixed-integer non-linear program (MINLP) aiming to minimize priority-weighted task completion latency while bounding aerial propulsion energy and enforcing strict collision safety constraints. To overcome the combinatorial action space, PA-MUJO decomposes the optimization into an analytical-RL hybrid architecture: optimal NOMA transmit powers are derived analytically via bisection search, while decentralized UAV actors learn continuous 3D velocity vectors and discrete scheduling associations through multi-agent proximal policy optimization (MAPPO) with centralized training and decentralized execution (CTDE).

Extensive empirical evaluations across 5×10^6 continuous environment steps on high-performance GPU infrastructure demonstrated that PA-MUJO achieves a **+45.7% improvement in episodic return** and a **45.4% reduction in task completion latency** compared to heuristic nearest-UAV Greedy Offloading. Furthermore, the 3-tier cooperative caching mechanism achieves a 77.4% **Air-to-Air hit rate**, significantly offloading traffic from terrestrial backhaul links while exhibiting robust resilience against adversarial ground jamming and Gauss-Markov ground user mobility. Future work will investigate online meta-reinforcement learning to adapt to non-stationary wind gust fields, explore decentralized multi-UAV coordination under imperfect channel state information (CSI), and examine cooperative aerial beamforming in massive MIMO-equipped UAV swarms.

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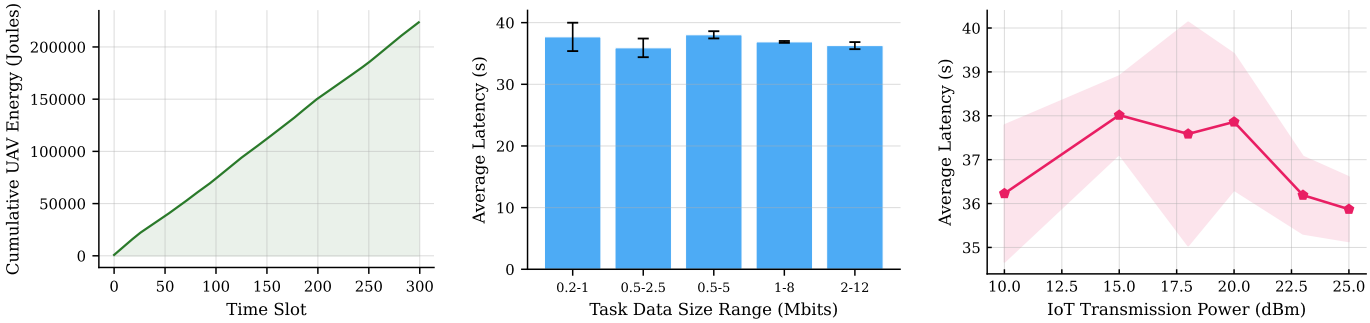
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(a) Cumulative Energy Consumption (b) Latency vs. Task Payload Size (c) Latency vs. IoT Transmit Power

Figure 9. Energy consumption and workload scalability: (a) cumulative UAV fleet energy profile, (b) average latency across task payload sizes (0.5 to 5.0 Mbit), and (c) latency response across IoT transmit powers (10 to 25 dBm).

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